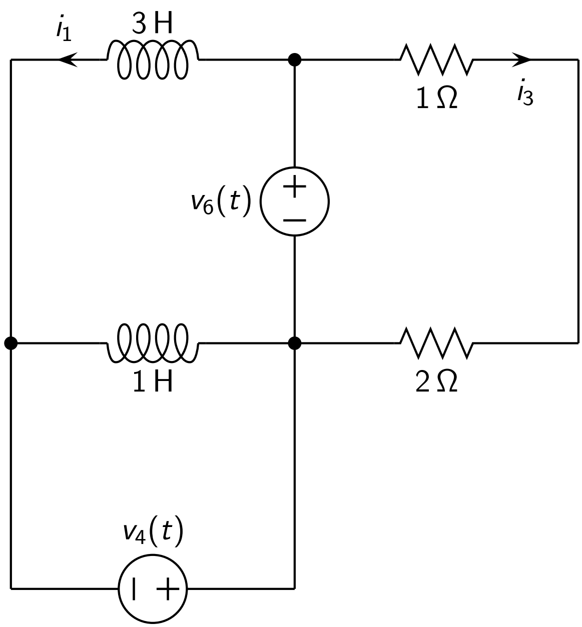
Problem: the circuit operates at steady-state. Find the expression of $i_1(t)$, $i_3(t)$.
Data: $v_4(t) = 3\cos(t - 135^o)\,\mathrm{V}$, $v_6(t) = 3\cos(t + 30^o)\,\mathrm{V}$.



Solution

$$i_1(t) = 0.261\cos(\,t - 143^o)\,\mathrm{A}$$

$$i_3(t) = \cos(\,t + 30^o)\,\mathrm{A}$$